

Assignment No-02

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Course : CSE 340

Section : 14.

Ans to the Q No-1

Multiplicand = $(13)_{10} = (01101)_2$

Multiplier = $(17)_{10} = (10001)_2$

number of iteration = 5

iteration	Multiplicand	Product (0000 10001)
1	01101	01101 10001 00110 11000
2	01101	00011 01100
3	01101	00001 10110
4	01101	00000 11011
5	01101	01101 11011 00110 11101

Verify:

$13 \times 17 = (221)_{10}$

Product = $(00110 11101)_2$
 $= (221)_{10}$

Ans to the Q No-2

ABB9609 = $\underbrace{1}_{\text{sign bit}} \underbrace{010 101}_{\text{exponent}} \underbrace{11011 1001 0110 0000 1001}_{\text{fraction}}$

bias = $2^{6-1} - 1 = 31$

bias exponent = $(010101)_2 = (21)_{10}$

actual exponent = $21 - 31 = -10$

fraction = $11011 1001 0110 0000 1001$

$= 2^{-1} + 2^{-2} + 2^{-4} + 2^{-5} + 2^{-6} + 2^{-9} + 2^{-11} + 2^{-12} + 2^{-18} + 2^{-21}$

$= 0.8620648384$

$$\text{Decimal number} = (-1)^{\text{sign bit}} \times (1 + \text{fraction}) \times 2^{\text{actual Exponent}}$$

$$= (-1) \times (1.8620648384 \times 2^{-10})$$

$$= -1.8620648384 \times 10^{-3}$$

(Ans)

Ans to the Q No-3

(a) $50.7896 + 79.83 - 29.58$

$$= 110010.11001 + 1001111.11010 - 11101.10010$$

$$= (1.1001011001 \times 2^5) + (1.00111111010 \times 2^6) - (1.110110010 \times 2^4)$$

$$= (0.11001011001 + 1.00111111010) \times 2^6 - (1.110110010 \times 2^4)$$

$$= (10.000001010011 - 0.01110110010) \times 2^6$$

$$= 1.10010100001 \times 2^6$$

(Ans)

(b) $64.2486 * 49.1832$

$$= 100\ 0000.00111 \times 110001.00101$$

$$= (1.00000000111 \times 2^6) \times (1.1000100101 \times 2^5)$$

$$= 1.10001\ 0101\ 0011\ 0000\ 0011 \times 2^{11}$$

(Ans)

Ans to the Q No-4

$$28.4810 - (-4.0210)$$

$$= 28.4810 + 4.0210$$

$$= 11100.01111 + 100.00000$$

$$= (1.110001111 \times 2^4) + (1.0000000 \times 2^2)$$

$$= (1.110001111 \times 2^4) + (0.010000000 \times 2^4)$$

$$= 10.000001111 \times 2^4$$

$$= 1.0000001111 \times 2^5$$

So actual exponent is 5.

$$\text{Now, bias} = 2^{8-1} - 1 = 127.$$

$$\therefore \text{Bias exp} = 5 + 127 = 132.$$

Range of bias exponent = 0 to $2^{8-1} - 1$. which means

1 to 255. As $1 < 132 < 255$. So there is no overflow

or underflow because the value is within range.

Ans to the Q No-5

(a) Bias is the number which is subtracted from a normal unsigned representation of a number's exponent to determine the real value and simplify it. The bias is added to the actual exponent in IEEE floating point representation to facilitate the encoding of both positive and negative exponents in an unsigned binary format. Adding bias gives us a range how big or small value we can represent.

(b) The optimized multiplication improves efficiency and performance by taking less space, which is reduced from 64 to 32 bits. It reduces computational complexity, memory requirements and execution time. In optimized method we can achieve faster execution by reducing the number of elementary operations like we do in long multiplication.

Ans to the Q No-6

1. fadda:

(i) Performs floating point addition of two single precision floating point numbers.

(ii) faddh $\rightarrow$ rd, rs1, rs2. [destination source1 source2]

(iii) Adds floating point numbers of source 1 and 2.
 $rd = rs1 + rs2$.

(iv) fadda $\rightarrow$ f0, f1, f2.

2. fsuba

(i) Performs floating point subtraction two single precision floating point numbers.

(ii) fsubh $\rightarrow$ rd, rs1, rs2. [destination source1 source2]

(iii) Subtracts floating point numbers of source 1 and 2.
 $rd = rs1 - rs2$.

(iv) fsuba $\rightarrow$ ~~f0, f1, f2~~ f0, f1, f2.

3. fmul.s:

i) Performs floating point multiplication of two single precision point numbers.

ii) $fmul.s \rightarrow rd, rs1, rs2$. [destination ~~named~~ source1 source2]

iii) Multiplies two floating point numbers and stores at destination.

$$fmul.s \rightarrow rd = rs1 * rs2$$

iv) $fmul.s$ ~~on~~ $fs, fs1, fs2$.

4. fdiv.s:

i) Performs floating point division of two single precision point numbers.

ii) $fdiv.s \rightarrow rd, rs1, rs2$. [destination source1 source2]

iii) It divides two floating point numbers and stores at destination.

$$fdiv.s \rightarrow rd = rs1 / rs2$$

iv) $fdiv.s$ ~~on~~ $fs, fs1, fs2$.

5. fsqrt.s:

i) Computes the square root of a single precision floating point number.

ii) $fsqrt.s \rightarrow rd, rs1$ [destination source1]

iii) If finds root of the source and stores it at destination.

$\text{fsgnrd} \rightarrow \text{rd} = \sqrt{\text{rs1}}$

iv fsgnrd fo, fa

6. fsgnd

i Compares two double precision floating point numbers for equality.

ii $\text{fsgnd} \rightarrow \text{rd}, \text{rs1}, \text{rs2}$ [destination source 1 source 2]

iii Values at source 1 and source 2 is compared, and stored in destination if ~~the~~ ^{equal then 1} ~~else~~ else store 0

iv fsgnd ^{rs1} ~~rs1~~ $\text{rs1}, \text{rs2}$

7. fble.d

i Checks if the double precision number is less than or equal to another.

ii $\text{fble.d} \rightarrow \text{rd}, \text{rs1}, \text{rs2}$ [destination source 1 source 2]

iii In the destination register if source 1 is less than source 2 the 1 or else store 0.

iv fble.d ^{rs1} ~~rs1~~ $\text{rs1}, \text{rs2}$

8. Slt r1

① Check if one single precision float point number is less than other.

[Destination source1 source2]

② Slt rd, rs1, rs2.

③ In the destination register if source 1 is less than source 2 then store 1 or else store 0.

④ Slt r1, \$0, \$1, \$2.

Ans to the Q No-7

Seq d. x5, \$1, \$2

beq x5, x0, jumpNotEqual.

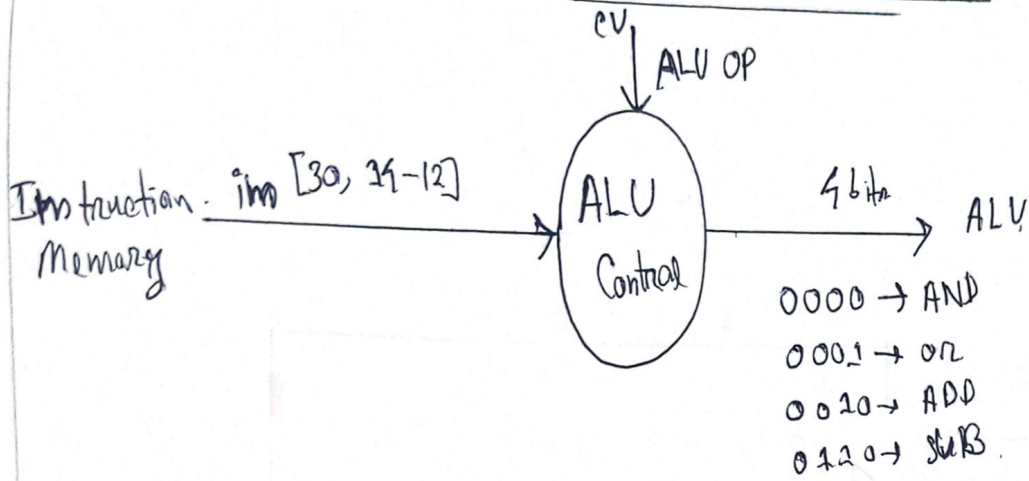
jumpEqual

exit:

~~jumpNotEqual~~

jumpNotEqual.

Ans to the Q No-8



(a) No, ALU control does not utilize bits 30 and 25-12 for load instructions. For LD ALU simply performs addition for base address and offset to calculate the effective memory address.

(b) ALU control utilizes instruction bits 30 and 25-12 used for R type and I type instructions. In R type function, function bit 30 makes the difference between ADD and SUB and function bit (25-12) determines the operation as ADD, AND, OR etc. For I type, function bit (25-12) specifies the immediate operation ADDI or AND. These bits uniquely determine the exact ALU operation needed.

ADD X22, X22, X23.

Ans to the Q No-9

